



National
Qualifications
2024

X857/76/22

Physics
Paper 1 — Relationships sheet

THURSDAY, 25 APRIL

9:00 AM – 9:45 AM



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Relationships required for Physics Higher

$$d = \bar{v}t$$

$$s = \bar{v}t$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd, \text{ or } W = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$P = \frac{E}{t}$$

$$p = mv$$

$$Ft = mv - mu$$

$$F = G \frac{m_1 m_2}{r^2}$$

$$t' = \frac{t}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

$$l' = l \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$f_o = f_s \left(\frac{v}{v \pm v_s} \right)$$

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}$$

$$z = \frac{v}{c}$$

$$v = H_0 d$$

$$W = QV$$

$$E = mc^2$$

$$I = \frac{P}{A}$$

$$I = \frac{k}{d^2}$$

$$I_1 d_1^2 = I_2 d_2^2$$

$$E = hf$$

$$E_k = hf - hf_0$$

$$v = f\lambda$$

$$E_2 - E_1 = hf$$

$$d \sin \theta = m\lambda$$

$$n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

$$\sin \theta_c = \frac{1}{n}$$

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

$$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$P = IV = I^2 R = \frac{V^2}{R}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$V_1 = \left(\frac{R_1}{R_1 + R_2} \right) V_S$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$E = V + Ir$$

$$C = \frac{Q}{V}$$

$$Q = It$$

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$\text{path difference} = m\lambda \quad \text{or} \quad \left(m + \frac{1}{2}\right)\lambda \quad \text{where } m = 0, 1, 2, \dots$$

$$\text{random uncertainty} = \frac{\text{max. value} - \text{min. value}}{\text{number of values}}$$

or

$$\Delta R = \frac{R_{\max} - R_{\min}}{n}$$

Additional relationships

Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Electron arrangements of elements

Group 1	Group 2
(1)	

(1)

Key	
Atomic number	
Symbol	
Electron arrangement	
Name	

Group 3 Group 4 Group 5 Group 6 Group 7 Group 0

(18)

2 He	2 He	(13)	(14)	(15)	(16)	(17)
2 He	Helium	5 B 2,3	6 C 2,4	7 N 2,5	8 O 2,6	9 F 2,7
10 Ne 2,8	Neon	Boron	Carbon	Nitrogen	Oxygen	Fluorine
18 Ar 2,8,8	Argon	13 Al 2,8,3	14 Si 2,8,4	15 P 2,8,5	16 S 2,8,6	17 Cl 2,8,7
36 Kr	Krypton	Aluminium	Silicon	Phosphorus	Sulfur	Chlorine
31 Ga	Gallium	32 Ge	33 As	34 Se	35 Br	36 Kr
2,8,18,3	Germanium	2,8,18,4	2,8,18,5	2,8,18,6	2,8,18,7	2,8,18,8
49 In 2,8,18, 18,3	Indium	50 Sn 2,8,18, 18,4	51 Sb 2,8,18, 18,5	52 Te 2,8,18, 18,6	53 I 2,8,18, 18,7	54 Xe 2,8,18, 18,8
81 Tl	Thallium	82 Pb	83 Bi	84 Po	85 At	86 Rn
2,8,18, 32,18,3	Lead	2,8,18, 32,18,4	2,8,18, 32,18,5	2,8,18, 32,18,6	2,8,18, 32,18,7	2,8,18, 32,18,8
			Bismuth	Polonium	Astatine	Radon

Transition elements

Sodium	Magnesium	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
2,8,8,1	2,8,8,2	2,8,9,2	2,8,10,2	2,8,11,2	2,8,13,1	2,8,13,2	2,8,14,2	2,8,15,2	2,8,16,2	2,8,18,1	2,8,18,2
Potassium	Calcium	Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
2,8,18,8,1	2,8,18,8,2	2,8,18,9,2	2,8,18,10,2	2,8,18,12,1	2,8,18,13,1	2,8,18,13,2	2,8,18,15,1	2,8,18,16,1	2,8,18,18,0	2,8,18,18,1	2,8,18,18,2
Rubidium	Strontium	Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg
2,8,18,18,8,1	2,8,18,18,8,2	2,8,18,18,9,2	2,8,18,32,10,2	2,8,18,32,11,2	2,8,18,32,12,2	2,8,18,32,13,2	2,8,18,32,14,2	2,8,18,32,15,2	2,8,18,32,17,1	2,8,18,32,18,1	2,8,18,32,18,2
Cesium	Barium	Lanthanum	Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum	Gold	Mercury
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn
2,8,18,32,18,8,1	2,8,18,32,18,8,2	2,8,18,32,18,9,2	2,8,18,32,32,10,2	2,8,18,32,32,11,2	2,8,18,32,32,12,2	2,8,18,32,32,13,2	2,8,18,32,32,14,2	2,8,18,32,32,15,2	2,8,18,32,32,17,1	2,8,18,32,32,18,1	2,8,18,32,32,18,2
Francium	Radium	Actinium	Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium

Lanthanides

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
2,8,18, 18,9,2 Lanthanum	2,8,18, 20,8,2 Cerium	2,8,18,21, 8,2 Praseodymium	2,8,18,22, 8,2 Neodymium	2,8,18,23, 8,2 Promethium	2,8,18,24, 8,2 Samarium	2,8,18,25, 8,2 Europium	2,8,18,25, 9,2 Gadolinium	2,8,18,27, 8,2 Terbium	2,8,18,28, 8,2 Dysprosium	2,8,18,29, 8,2 Holmium	2,8,18,30, 8,2 Erbium	2,8,18,31, 8,2 Thulium	2,8,18,32, 8,2 Ytterbium	2,8,18,32, 9,2 Lutetium

Actinides

89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
2,8,18,32, 18,9,2	2,8,18,32, 18,10,2	2,8,18,32, 20,9,2	2,8,18,32, 21,9,2	2,8,18,32, 22,9,2	2,8,18,32, 24,8,2	2,8,18,32, 25,8,2	2,8,18,32, 25,9,2	2,8,18,32, 27,8,2	2,8,18,32, 28,8,2	2,8,18,32, 29,8,2	2,8,18,32, 30,8,2	2,8,18,32, 31,8,2	2,8,18,32, 32,8,2	2,8,18,32, 32,9,2
Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium